

Sri Lanka Institute of Information Technology



Hotel Booking & Management System Final Report ITP-19

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DECLARATION

We hereby declare that this report is our own work and has not been submitted in part or full to any other university or institution for any other purpose.

We confirm that all sources of information used in this report have been properly acknowledged and referenced.

We also declare that all group members have contributed fairly and equally to the completion of this project.

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ABSTRACT

The Hotel Booking and Management System is a web-based application developed to address common challenges faced by small and medium-scale hotels in managing reservations, customer interactions, and operational processes. Traditional systems often rely on manual or semi-automated methods, leading to inefficiencies such as double bookings, delayed confirmations, and poor data management.

This project introduces an integrated solution that automates hotel operations, including room booking, event and hall management, payment processing, and transport services. Additionally, the system incorporates advanced features such as an AI-powered chatbot for customer support and an operational analytics center that provides predictive insights and automated reporting.

The system was designed using modern web technologies, ensuring scalability, usability, and efficiency. The frontend provides an interactive user experience, while the backend ensures secure data handling and real-time processing.

The implementation of this system improves operational efficiency, enhances customer experience, and reduces human errors. The results demonstrate that the system successfully meets its objectives by providing a reliable and intelligent platform for hotel management.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
UI	User Interface
UX	User Experience
DB	Database
API	Application Programming Interface
ER	Entity Relationship
UML	Unified Modeling Language
CRUD	Create, Read, Update, Delete

Chapter 1: Introduction

1.1 Problem Statement

In the modern hospitality industry, travelers increasingly rely on online platforms to search, compare, and book accommodations across multiple hotels. However, many existing solutions either focus on individual hotel management or lack comprehensive features that support both customers and hotel providers within a unified system. This creates a gap in the market for a centralized platform that efficiently connects multiple hotels with potential customers.

One of the major challenges faced by users is the difficulty in comparing different hotels based on availability, pricing, location, and services in real time. Without a centralized system, customers often need to visit multiple websites or rely on outdated information, leading to inconvenience and poor decision-making. Additionally, inconsistencies in booking processes across different hotels can result in confusion, delays, and booking errors.

From the perspective of hotel providers, especially small and medium-scale establishments, there is a lack of accessible platforms that allow them to manage listings, update availability, and reach a wider audience effectively. Many hotels do not have the resources to develop their own advanced booking systems, limiting their visibility and competitiveness in the market.

Furthermore, existing platforms often lack advanced features such as integrated analytics, intelligent pricing strategies, and automated customer support. This limits the ability of both customers and hotel administrators to make informed decisions and optimize operations.

Therefore, there is a clear need for a multi-hotel booking platform that provides a centralized, efficient, and intelligent solution. Such a system should enable users to easily search and compare hotels, while allowing hotel providers to manage their services effectively, ultimately improving user experience and operational efficiency across the hospitality ecosystem.

1.2 Motivation

The motivation behind this project comes from the growing demand for digital travel platforms that simplify the hotel booking experience for customers while also supporting hotel providers with efficient management tools. Modern users expect to compare multiple hotels in one place, view accurate room availability, and complete bookings quickly and conveniently. At the same time, hotel providers need a reliable platform to promote their properties, manage reservations, and improve operational efficiency.

This project is motivated by the idea of developing a platform similar to Booking.com, but enhanced with intelligent and management-oriented features. Instead of focusing only on room reservations, the system combines hotel discovery, booking, payment processing, transport support, hall booking, AI-based customer communication, and operational analytics into one centralized solution.

The project also explores how artificial intelligence can add value to hotel booking platforms. Features such as chatbot assistance, anomaly detection, predictive analytics, and dynamic pricing can improve both user experience and business decision-making. These capabilities make the system more than a standard booking website and position it as a smart hospitality platform.

Overall, the project is motivated by the need to provide a modern, scalable, and intelligent booking solution that benefits both travelers and hotel providers.

1.3 Aim and Objectives

Aim

The main aim of this project is to develop a centralized multi-hotel booking and management platform that enables customers to search, compare, and book hotels efficiently while providing intelligent management and decision-support features for hotel providers.

Objectives

The objectives of the system are as follows:

- To develop a web-based platform that allows users to search and compare hotels by location, dates, pricing, and availability
- To implement a room booking mechanism with real-time availability checking
- To provide hotel administrators with tools to manage hotels, room types, and booking-related information
- To integrate a payment, billing, and invoice management component for booking transactions
- To provide additional services such as transport requests and event hall booking
- To implement an AI-powered chatbot to support customer communication and reservation-related inquiries
- To develop an analytics and reporting component with predictive and anomaly-detection capabilities
- To enhance user convenience, operational efficiency, and service quality through automation and intelligent features

1.4 Literature Review

Online hotel booking platforms have significantly transformed the hospitality industry by allowing users to search, compare, and reserve accommodations through digital channels. Among the most recognized systems, Booking.com is widely used for its ability to provide access to multiple hotels, real-time availability, customer reviews, and pricing comparisons. It offers users a convenient way to evaluate different accommodation options before making a reservation.

Airbnb is another major platform in the accommodation sector. Unlike traditional hotel booking systems, Airbnb focuses on connecting hosts and guests for short-term stays. It provides flexible booking options and a user-friendly interface, but its primary focus is on property-sharing rather than comprehensive hotel-oriented operational support.

Many conventional hotel management systems are designed to support the internal operations of a single hotel, such as room allocation, customer records, and billing. While such systems are useful for local administration, they generally do not function as centralized marketplaces where customers can compare and book across multiple hotels. They also often lack intelligent features such as predictive analytics, AI-based assistance, and smart operational monitoring.

The proposed system differs from these approaches by combining both customer-facing and provider-facing functionalities within a single platform. Similar to Booking.com, it enables users to discover and compare multiple hotels. At the same time, it extends beyond a typical booking marketplace by including hotel-side management features, event and hall booking, guest transport services, AI chatbot interaction, operational analytics, and dynamic pricing support. This makes the platform a more comprehensive and intelligent solution for the hospitality domain.

1.5 Solution Overview

The proposed system is a web-based multi-hotel booking and management platform designed to connect travelers with hotel providers through a centralized digital environment. The system allows users to search hotels by location and dates, compare room options, view availability, and complete bookings through an interactive interface. This supports the core concept of an online booking marketplace similar to Booking.com.

In addition to standard room booking functionality, the platform includes several extended modules. These include event and hall booking, guest transport request handling, payment and invoice processing, AI-powered customer support, and operational analytics. Hotel administrators are able to manage hotel and room details, monitor performance, respond to booking-related activities, and use system-generated insights to support decision-making.

The inclusion of AI-based features such as chatbot support, anomaly detection, predictive reporting, and dynamic pricing makes the system more advanced than a traditional booking platform. Therefore, the proposed solution serves both as a customer-oriented hotel discovery and booking platform and as a smart operational support system for participating hotels.

GitHub Repository: <https://github.com/rplk3/StayFlow.git>

Chapter 2: Requirement Analysis

2.1 Introduction

This chapter describes the process of identifying, analyzing, and structuring the system requirements for the proposed multi-hotel booking and management platform. The analysis focuses on understanding the needs of different stakeholders, evaluating the feasibility of the system, and defining both functional and non-functional requirements.

2.2 Stakeholder Analysis

The system involves multiple stakeholders, each with different roles and expectations. Identifying these stakeholders is essential to ensure that the system meets all user needs effectively.

Stakeholder	Description	Responsibilities
Customers (Guests)	Users who search and book hotels	Search hotels, compare options, make bookings, request services
Hotel Providers (Admins)	Hotel owners or staff managing listings	Add/update hotel details, manage rooms, monitor bookings
Platform Administrator	System-level controller	Manage users, monitor system, ensure security and reliability
Payment Gateway	External service for transactions	Process payments securely
System Developers	Development team	Design, implement, and maintain the system

2.3 Feasibility Analysis

2.3.1 Technical Feasibility

The technical feasibility assessment evaluates whether the team has the necessary technologies, tools, and skills to successfully build the proposed system.

Factor	Assessment	Verdict
Frontend Development	React.js or similar framework well within the team's IT skill set	Feasible
Backend Development	Node.js / Express or Spring Boot standard for web-based systems	Feasible
Database	MySQL or MongoDB for storing bookings, guests, rooms, payments	Feasible
AI / ML Integration	Generative AI for analytics (OpenAI API or similar), NLP chatbot using Dialogflow or Rasa	Feasible with care
Payment Gateway	Mock/sandbox gateway (e.g., Stripe test mode) no live payment processing required	Feasible
Dynamic Pricing Logic	Rule-based pricing for weekends/seasons implementable without complex ML	Feasible
Email Notifications	SMTP/email service (e.g., NodeMailer, SendGrid) for booking confirmations	Feasible
Hosting & Deployment	Cloud hosting via Vercel, Netlify (frontend) and Render or AWS free tier (backend)	Feasible

2.3.2 Economic Feasibility

The economic feasibility study evaluates the costs involved in developing the system against the benefits it would deliver. Since this is an academic project, no commercial budget is allocated; however, a realistic cost estimate is presented below.

Cost Category	Estimated Cost (LKR)	Notes
Development Tools & IDEs	0	Free tools (VS Code, IntelliJ Community)
Hosting (Frontend)	0	Vercel / Netlify free tier
Hosting (Backend)	0 – 3,000/month	Render free tier or low-cost plan
Database Hosting	0	MongoDB Atlas free tier
AI API Usage (OpenAI etc.)	2,000 – 5,000	Limited usage for academic purposes
Email Service (SendGrid)	0	Free tier (up to 100 emails/day)
Domain Name (optional)	2,500/year	Optional for demonstration
Total Estimated Cost	4,500 – 10,500 LKR	Minimal academic cost

2.3.3 Operational Feasibility

Operational feasibility examines whether the system will be accepted and effectively used by its intended users once deployed.

Stakeholder	Role in System	Operational Concern	Assessment
Hotel Admin	Manage rooms, bookings, transport, pricing	Needs a clear dashboard; no specialized technical knowledge assumed	Addressed via admin panel
Hotel Staff	Handle day-to-day operations	Simple UI with guided workflows	Feasible
Guests / Customers	Search rooms, book, pay, request transport	Self-service portal with email confirmations	User-friendly design planned
System (AI Module)	Auto-generate reports, detect anomalies	Operates in background; minimal user intervention	Automated

2.4 SWOT Analysis

Strengths	Weaknesses
• Centralized multi-hotel platform • Integration of AI features	• Complexity due to multiple modules • Initial learning curve for users
• Real-time booking and availability • Automated processes reduce errors	• Dependency on internet connectivity • System maintenance required
Opportunities	Threats
• Growing demand for online booking platforms • Expansion to other services (travel, tourism)	• Competition from existing platforms (Booking.com, Airbnb) • Security risks and cyber threats
• Integration with third-party services	• Rapid technology changes

2.5 Functional Requirements

Functional requirements define the core operations of the system.

2.5.1 Customer Requirements

- Search hotels based on location and dates
- View hotel details, pricing, and availability
- Compare different hotels
- Book rooms and receive confirmation
- Make secure online payments
- Request additional services (transport, events)
- Interact with AI chatbot for support

2.5.2 Hotel Provider Requirements

- Add, update, and remove hotel listings
- Manage room types, pricing, and availability
- View and manage bookings
- Approve or reject requests (e.g., event bookings)
- Monitor performance through analytics

2.5.3 System Requirements

- Handle user authentication and authorization
- Process payments and generate invoices
- Send notifications (email confirmations, updates)
- Maintain booking records and history
- Generate reports and analytics

2.6 Non-Functional Requirements

Non-functional requirements define system quality attributes.

2.6.1 Performance

- System should respond quickly to user requests
- Booking operations should be processed in real time

2.6.2 Security

- Secure user authentication and data protection
- Safe payment processing using encryption

2.6.3 Usability

- User-friendly interface
- Easy navigation for both customers and admins

2.6.4 Scalability

- System should support increasing number of users and hotels

2.6.5 Reliability

- System should minimize downtime and errors
- Ensure accurate booking and data consistency

2.7 Requirements Modeling

To better understand system functionality, requirement models are used.

2.7.1 Use Case Overview

The system includes the following main use cases:

- Search Hotels
- View Hotel Details
- Book Room
- Process Payment
- Manage Hotels
- Generate Reports
- Chat with AI Assistant

These use cases represent interactions between users and the system.

2.7.2 System Actors

- Customer
- Hotel Provider
- Admin

Each actor interacts with the system based on their role and permissions.

2.8 Summary

This chapter presented a comprehensive analysis of system requirements, including stakeholder identification, feasibility evaluation, SWOT analysis, and requirement modeling. These requirements form the foundation for the system design and development discussed in the next chapter.

Chapter 3: Design and Development

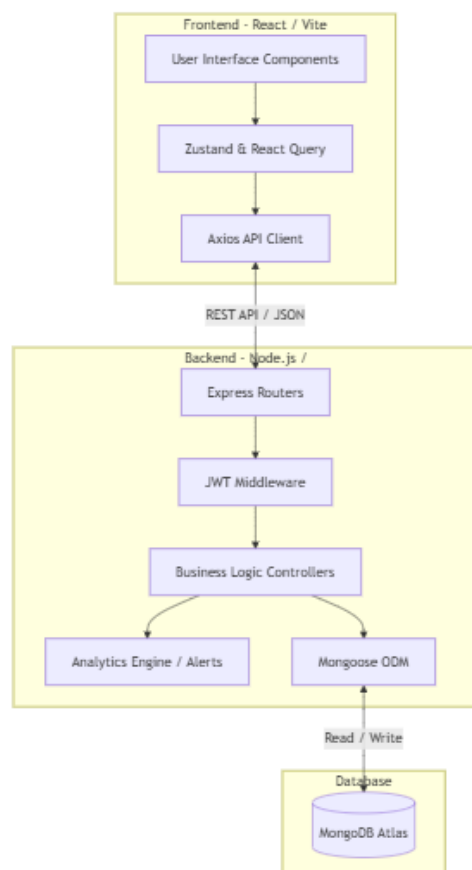
3.1 Introduction

This chapter presents the design and development of the proposed multi-hotel booking and management system. It explains how the system is structured, how different components interact, and how the system was implemented to meet the requirements identified in Chapter 2. The chapter includes architectural design, system workflows, database structure, and development details.

3.2 System Architecture

The system follows a multi-tier architecture consisting of the frontend, backend, and database layers. The frontend handles user interaction, while the backend processes business logic and communicates with the database. External services such as payment gateways and AI services are also integrated into the system.

The frontend is responsible for displaying hotel listings, handling user inputs, and providing an interactive interface for customers and administrators. The backend manages booking logic, authentication, payment processing, and communication between components. The database stores user data, hotel details, bookings, and transaction records.



The architecture ensures separation of concerns, scalability, and efficient data processing. It also allows easy integration of additional features such as AI modules and third-party services.

3.3 System Components

The system is divided into several major components, each responsible for a specific functionality.

3.3.1 Hotel Search and Booking Component

This component allows users to search for hotels based on location and dates. It retrieves available hotels and room details in real time and enables users to compare options before making a booking.

3.3.2 Hotel Management Component

This component is used by hotel providers to manage hotel listings, room types, pricing, and availability. It allows administrators to update information and monitor bookings efficiently.

3.3.3 Payment and Billing Component

This component handles secure payment processing, invoice generation, and payment tracking. It ensures that all transactions are processed reliably and securely.

3.3.4 Transport and Event Management Component

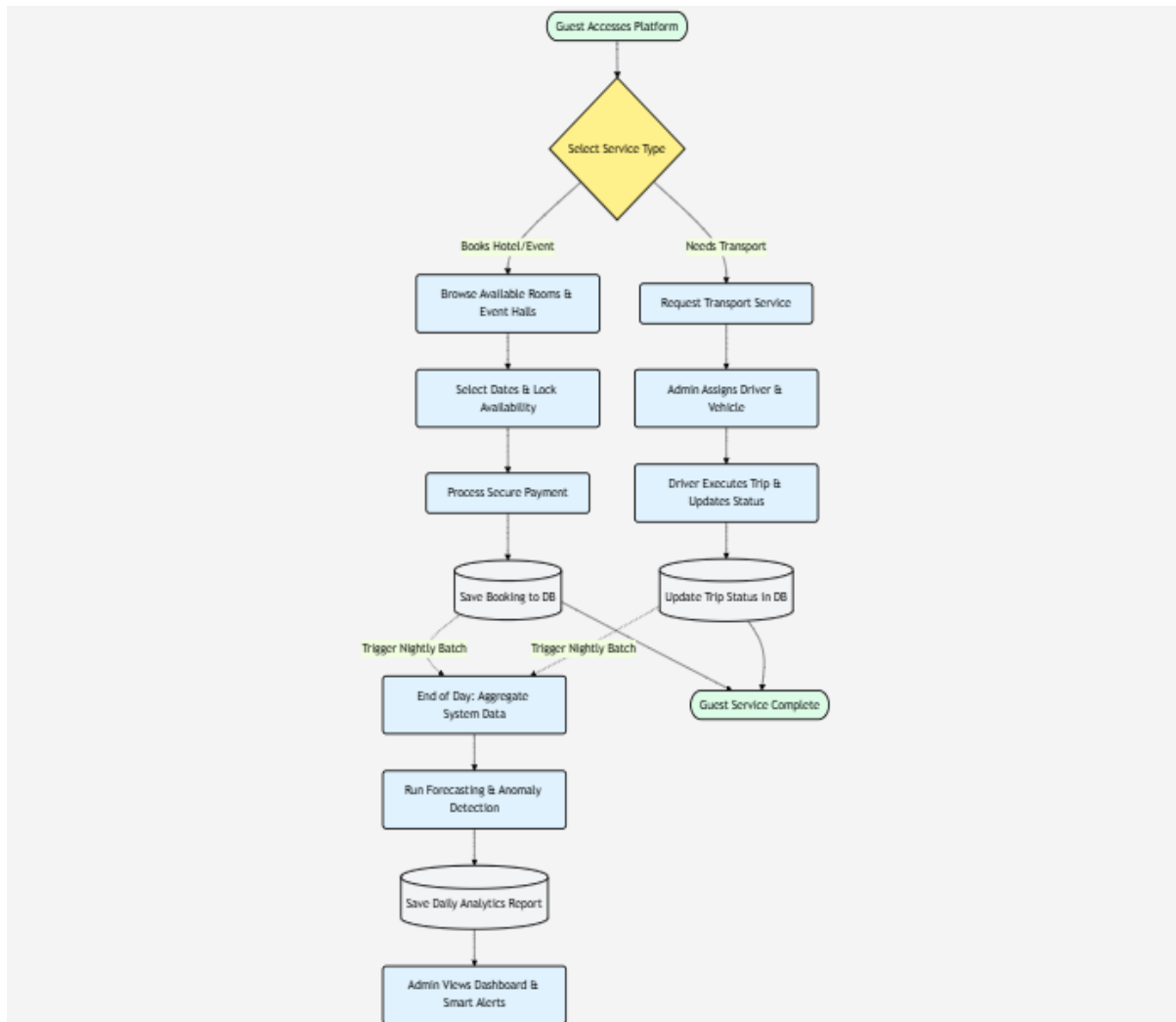
This module allows users to request transport services and book event halls. It links additional services with the main booking system to provide a complete experience.

3.3.5 AI-Based Components

The system includes an AI chatbot for customer interaction and an analytics module for reporting and prediction. These components enhance system intelligence and improve decision-making.

3.4 System Workflow

The system workflow describes how users interact with the system from searching for hotels to completing a booking.



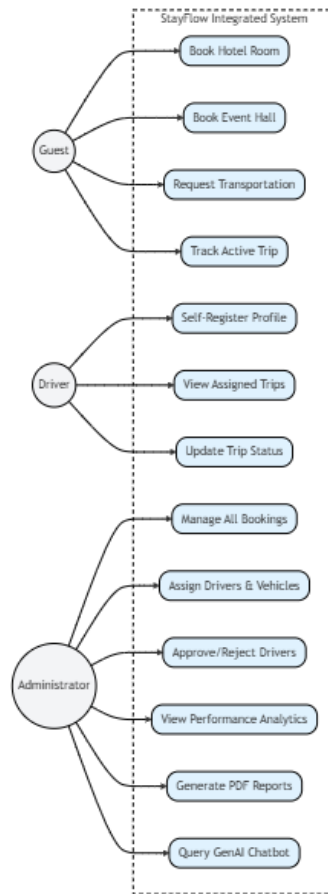
A typical workflow includes:

1. User searches for hotels based on location and dates
2. System retrieves and displays available options
3. User selects a hotel and room
4. User proceeds with booking and payment
5. System confirms booking and updates availability
6. Notifications are sent to the user

This workflow ensures a smooth and efficient booking process.

3.5 Use Case Diagram

The use case diagram illustrates the interactions between system actors and the system functionalities.



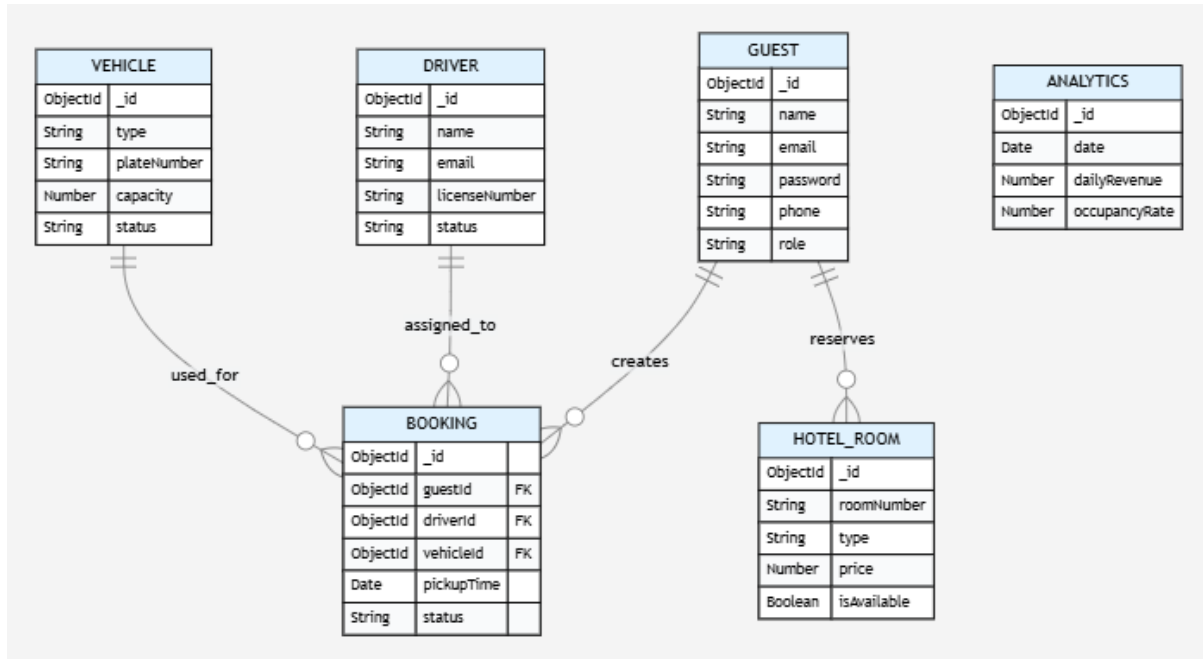
The main actors include:

- Customer
- Hotel Provider
- System Administrator

Each actor performs specific actions such as booking rooms, managing hotels, and monitoring system activities.

3.6 Database Design

The database is designed to store and manage all system data efficiently. It includes multiple entities such as users, hotels, rooms, bookings, and payments.



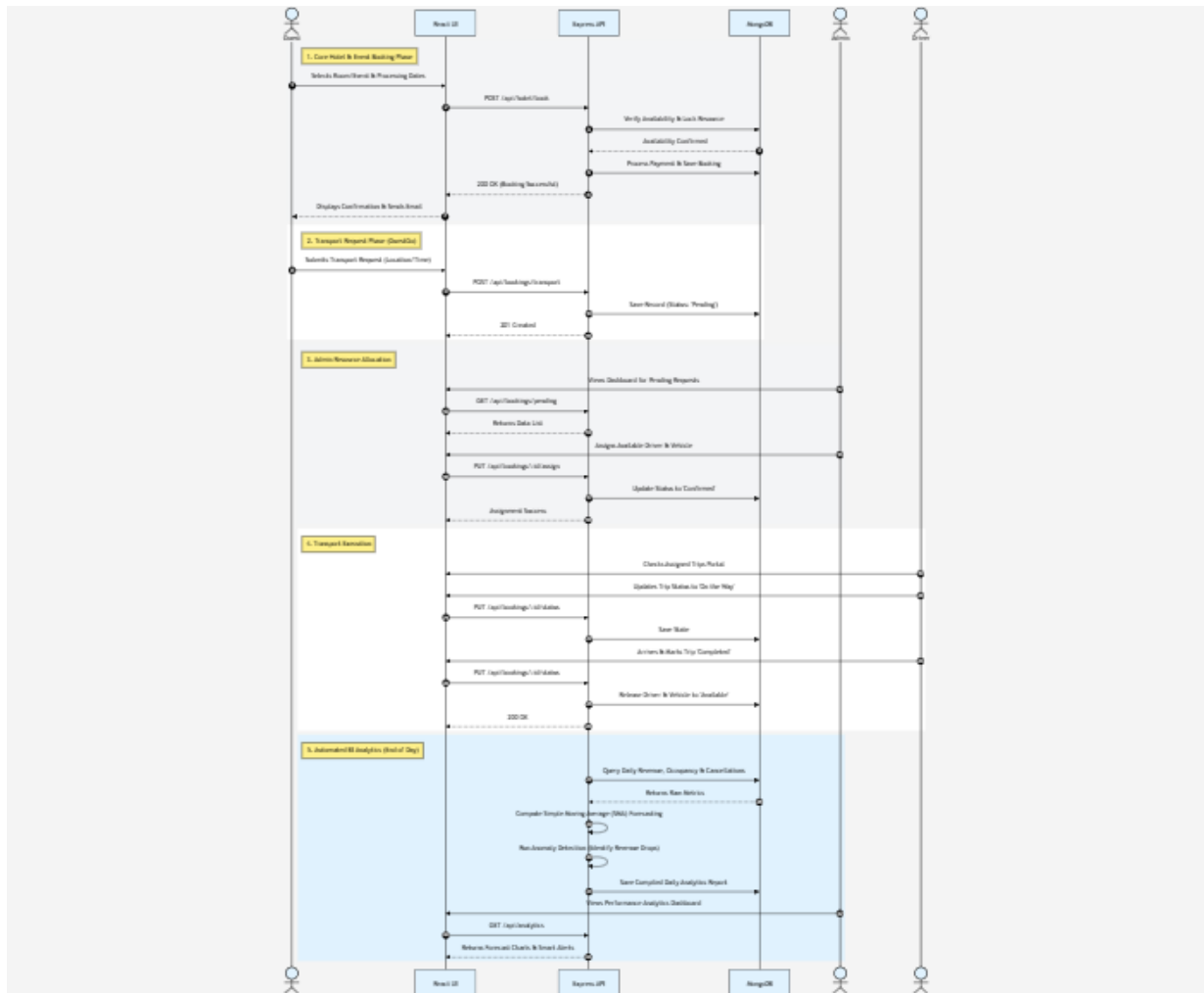
Key entities include:

- Vehicle (Guest transporting vehicle details)
- Driver (Guest transporting driver details)
- Hotels (hotel details and location)
- Rooms (room types, pricing, availability)
- Booking (reservation details)
- Analytics (Performance analysis of system)

The relationships between these entities ensure data consistency and efficient query processing.

3.7 Sequence Diagram

The sequence diagram shows how different components interact during the booking process.



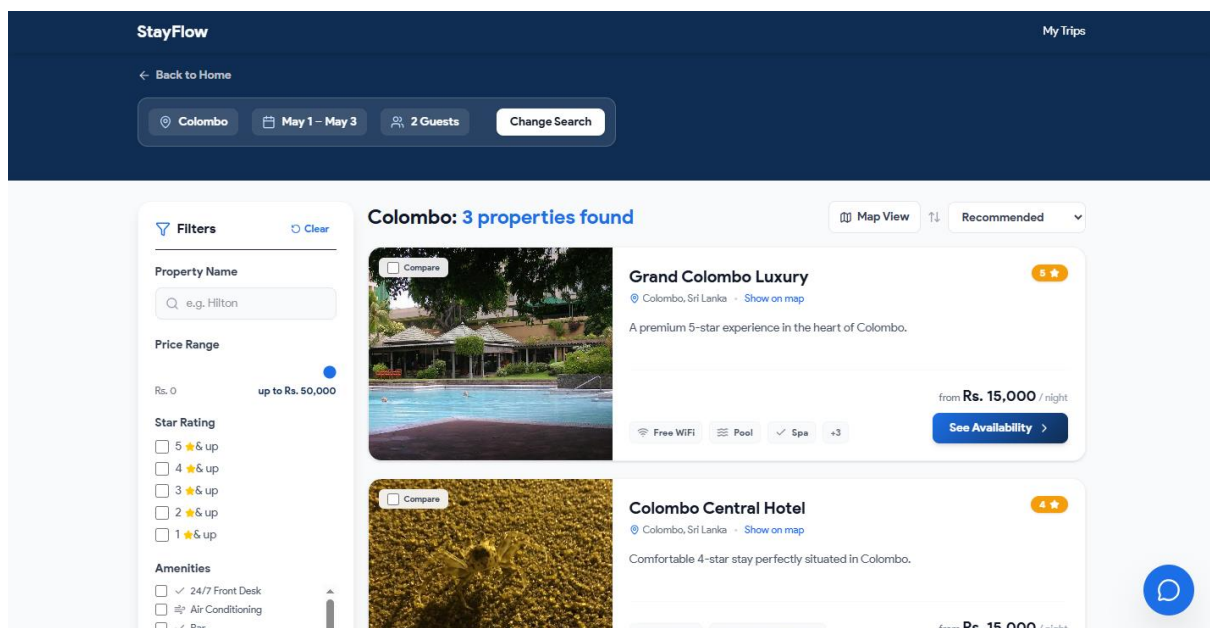
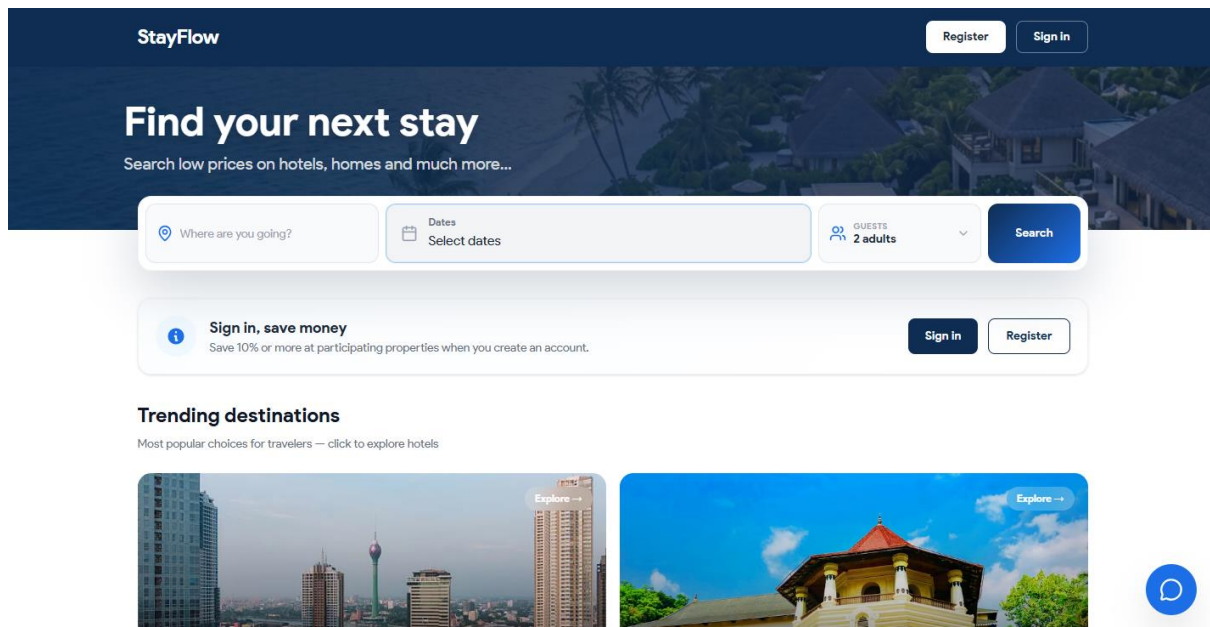
The sequence typically includes:

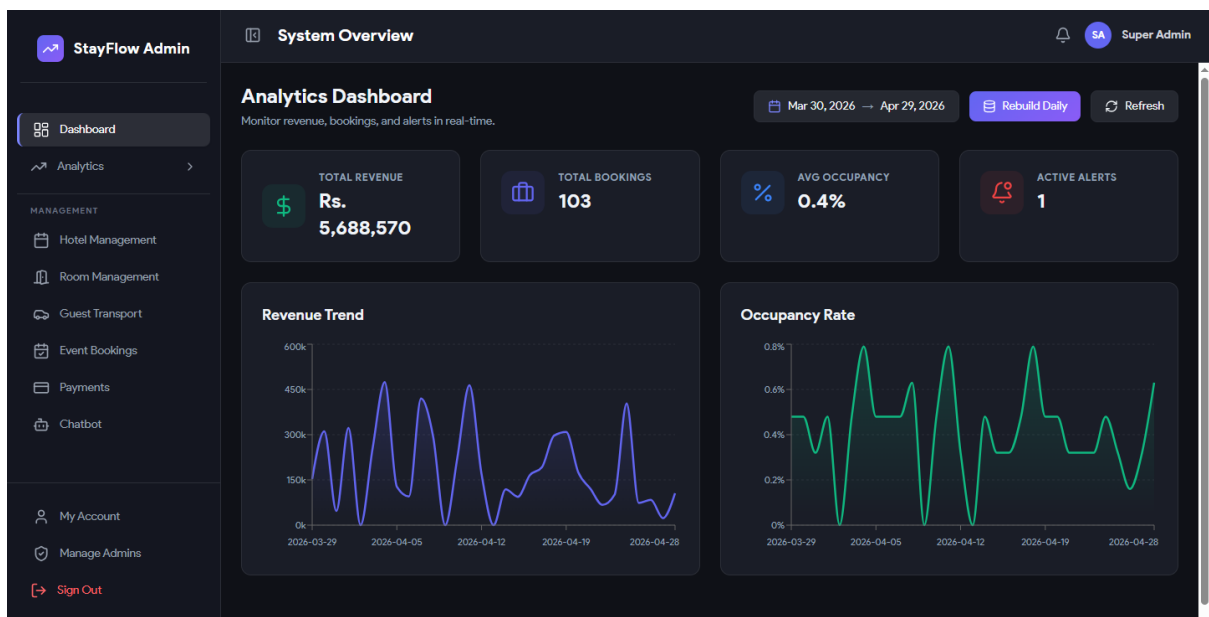
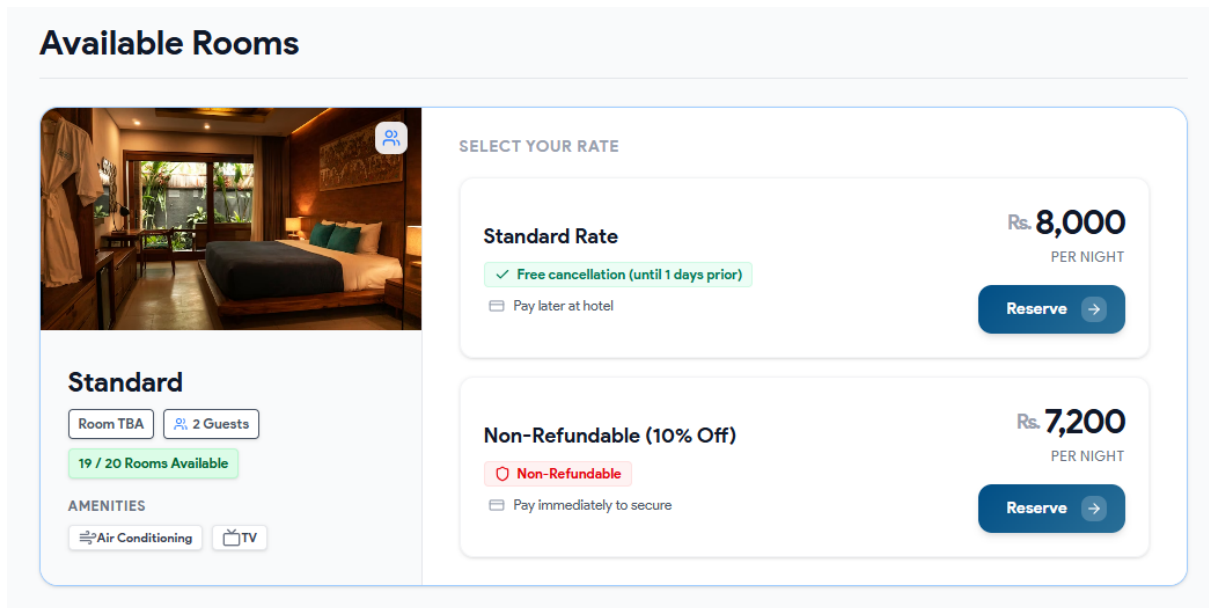
- User sends booking request
- System validates availability
- Payment is processed
- Booking is confirmed
- Database is updated

This interaction ensures that all operations are completed in a correct and logical order.

3.8 User Interface Design

The user interface is designed to be simple, intuitive, and user-friendly. It provides easy navigation and clear presentation of information for both customers and administrators.





Key interfaces include:

- Home page with search functionality
- Hotel listing and comparison page
- Booking and payment interface
- Admin dashboard for management

The design focuses on usability and accessibility to enhance user experience.

3.9 Development Tools and Technologies

The system was developed using modern technologies to ensure performance and scalability.

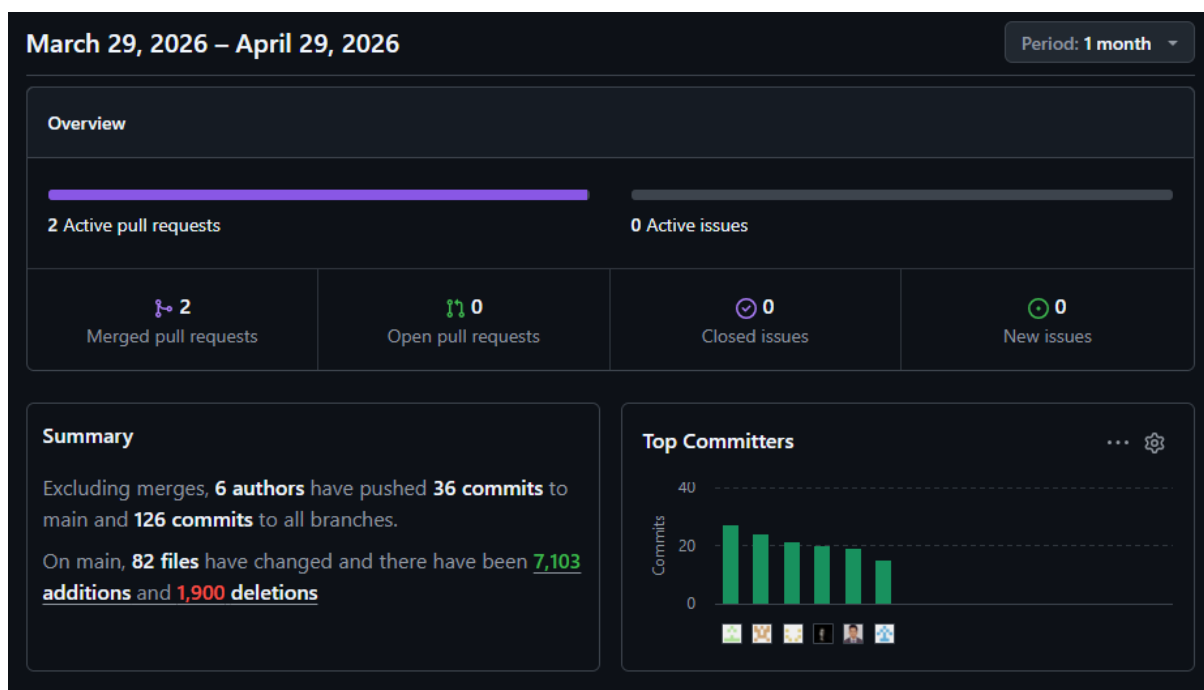
- Frontend: **React**
- Backend: **Express.js**
- Database: **MongoDB**
- Version Control: **GitHub**

These tools enable efficient development, testing, and deployment of the system.

3.10 Version Control and Development Evidence

Version control was maintained throughout the project using Git and GitHub. The repository was used to manage source code, track changes, and support collaboration among team members.

Each member contributed to the project through commits, which represent the development progress over time. The commit history shows the implementation of different system components, bug fixes, and feature updates.



The use of GitHub ensured proper collaboration, version control, and transparency in the development process.

GitHub Repository: <https://github.com/rplk3/StayFlow.git>

Chapter 4: Results and Evaluation

4.1 Introduction

This chapter presents the outcomes of the developed multi-hotel booking and management system and evaluates its performance based on the defined requirements. The evaluation includes system functionality, testing results, performance analysis, and user feedback.

4.2 System Implementation Results

The developed system successfully provides a centralized platform for users to search, compare, and book hotels. The system integrates multiple functionalities including booking management, payment processing, transport services, event management, and AI-based features.

The implementation demonstrates that users can interact with the system efficiently through a user-friendly interface. Hotel providers are able to manage their listings, update room availability, and monitor bookings effectively.

Key results include:

- Successful hotel search and filtering based on location and dates
- Real-time room availability updates
- Secure booking and payment processing
- Integration of additional services such as transport and event booking
- AI chatbot interaction for customer support
- Automated reporting and analytics generation

4.3 Functional Testing

Functional testing was conducted to verify that all system features operate according to the requirements.

Sample test cases include:

- Hotel search functionality
- Booking process validation
- Payment processing
- Admin management features
- Chatbot response accuracy

All major functionalities were tested and confirmed to work as expected.

4.4 Performance Evaluation

The system performance was evaluated based on responsiveness and efficiency.

Key observations:

- The system responds quickly to user queries and searches
- Booking operations are processed in real time
- Database operations are optimized for fast data retrieval
- The system performs efficiently under normal usage conditions

4.5 User Feedback

User feedback was collected to evaluate system usability and satisfaction.

Common feedback includes:

- The system is easy to use and navigate
- Booking process is fast and efficient
- Interface is clean and user-friendly
- AI chatbot provides helpful responses

This feedback indicates that the system meets user expectations.

4.6 Discussion

The system successfully meets the objectives defined in Chapter 1 and the requirements outlined in Chapter 2. The integration of multiple modules into a single platform improves overall efficiency and user experience.

Compared to traditional systems, the proposed platform provides better accessibility, automation, and intelligent features. The addition of AI components further enhances the system's capability to support decision-making and customer interaction.

However, some limitations exist, such as dependency on internet connectivity and the need for further optimization for large-scale deployment.

4.7 Summary

This chapter evaluated the developed system based on functionality, performance, and user feedback. The results confirm that the system operates effectively and fulfills its intended purpose. The next chapter provides the final conclusion of the project.

Chapter 5: Conclusion

5.1 Conclusion

This project successfully developed a centralized multi-hotel booking and management platform that allows users to search, compare, and book hotels efficiently. The system addresses the limitations of traditional booking methods by providing a unified platform where customers can access real-time information about hotel availability, pricing, and services.

The implementation of key features such as room booking, event and hall management, transport services, payment processing, and AI-based components demonstrates the system's ability to handle multiple aspects of hotel operations within a single solution. The inclusion of intelligent features such as chatbot support and operational analytics further enhances the functionality of the system.

All the objectives defined at the beginning of the project were achieved. The system provides a user-friendly interface for customers and effective management tools for hotel providers. It reduces booking errors, improves efficiency, and enhances overall user experience.

5.2 Future Improvements

Although the system meets its current objectives, there are several areas for future improvement.

- Integration with real-world payment gateways for live transactions
- Expansion to include more travel-related services such as flight bookings and tour packages
- Enhancement of AI capabilities for better recommendations and personalization
- Mobile application development for improved accessibility
- Scalability improvements to support large-scale deployment

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Appendix A: Contribution of Team Members

Member Name	Student ID	Contribution	Contribution Description
Wijesinghe W.P.H	IT24103110	17%	Event & Hall Management
Rashmindika N.H.A.R	IT23320482	16%	Payment & Billing System
Bambaragama H.M.G.K	IT24102523	17%	Transport Management
Mandinu W.A.C	IT24100648	17%	AI Chatbot
Perera M.R.R	IT24100688	17%	Analytics System
Perera G.A.R	IT24101162	16%	Room Management